The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Canonical Minimal Visible-Response
Representative Necessity / Uniqueness Theorem

Abstract

The current active-primary theorem already proved that the full canonical current-body image core collapses benchmark-facingly to the canonical minimal visible-response representative $\mathfrak{S}^{\mathrm{disp,resp,red,min}}$ on the fixed section-centered displacement body. The current benchmark criterion therefore made the honest next move explicit: either derive that visible-response datum from still more primitive explicit substrate structure, or prove a sharper theorem strictly inside that datum.

The present note takes that honest internal route. It proves that on the recorded section-centered hidden-response route the canonical minimal visible-response representative is already necessary and unique at benchmark scope up to visible-response-faithful equivalence. More precisely, any benchmark-faithful reduction that preserves the fixed proto-section benchmark base, preserves the current visible-response route, and preserves the benchmark one-metric observer package must determine the same section-centered displacement-body datum together with the same positive visible-response operator on that body. Conversely, nothing beyond that datum is needed at current benchmark scope, because the earlier current-image-core collapse theorem already shows that the current body-response core and every currently recorded downstream benchmark-facing consequence factor through $\mathfrak{S}^{\mathrm{disp,resp,red,min}}$.

This is not yet a completed first-principles derivation of GR from substrate microphysics. Its positive result is narrower and more honest. It removes the remaining presentational freedom at the current visible-response frontier on the recorded route. The next primary manuscript must therefore derive or force that current visible-response datum itself from still more primitive explicit substrate structure. Another same-output deeper relay or a re-expansion back to the full current image-core presentation, the larger reduced-carrier presentation, the older projection/body presentation, or the full displacement-response substrate law is no longer the right front-line move.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity now has a sharper task than merely moving to another larger realization of the same visible-response data. The current canonical-current-body-image-core collapse theorem already spent the full image-core presentation as the primitive stopping point on the active section-centered hidden-response route. Once the fixed section-centered displacement body together with the positive visible-response operator are held fixed, the current body-response core and every currently recorded downstream benchmark-facing consequence follow unchanged. The present frontier is therefore already internal to that visible-response datum itself.

That status leaves only two honest options. One can derive or justify the current visible-response datum from still more primitive explicit substrate structure in a way that genuinely changes benchmark-facing status. Or, if that deeper forcing law is not yet cleanly available, one can prove a sharper internal theorem that removes the remaining presentational freedom strictly inside the visible-response datum. The present note takes the second route. Its claim is that, on the currently recorded route, the canonical minimal visible-response representative is already the unique minimal benchmark-facing datum up to visible-response-faithful equivalence.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen the larger current-image-core, reduced-carrier, projection/body, displacement-response-substrate-law, pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations as if they were again independently live. Its narrower benchmark-facing role is to prove that the current visible-response datum is already the unique minimal datum of the recorded route at current scope.

2. Fixed Inputs from the Current Visible-Response Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded visible-response frontier and current route)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base, in particular the recorded proto-section body* $\mathcal{B}_\sigma^{\mathrm{psec}}$ *and shell* $\mathcal{Z}_\sigma^{\mathrm{psec}}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{ppprese c}}$$

   whose fiber

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}}(y) \equiv \{v \in V_\sigma^{\mathrm{ppprese c}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}\}$$

   is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

3. *the current positive visible-response operator*

$$S_\sigma^{\mathrm{disp,resp,vis}}(y) : V_\sigma^{\mathrm{ppprese c}} \to V_\sigma^{\mathrm{ppprese c}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

   smooth in y *and positive definite on every fiber;*

4. *the induced current body-response route*

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}} : \mathcal{D}_\sigma^{\mathrm{ppprese c}} \to [0, \infty), \qquad \widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y, v) = \frac{1}{2}\left\langle v, S_\sigma^{\mathrm{disp,resp,vis}}(y) v \right\rangle;$$

5. *the current canonical minimal visible-response representative*

$$\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}} \equiv \left(V_\sigma^{\mathrm{ppprese c}}, \mathrm{id}_{V_\sigma^{\mathrm{ppprese c}}}, S_\sigma^{\mathrm{disp,resp,vis}}\right)$$

   over the fixed displacement body $\mathcal{D}_\sigma^{\mathrm{ppprese c}}$*;*

6. *the current downstream continuation, namely the current body-response core together with*

$$\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyop}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb}},$$

$$\mathfrak{R}^{\mathrm{sub,proto,prese c,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,prese c}},$$
$$\mathfrak{R}^{\mathrm{sub,proto,prese c,sec}},$$

   and the current continuity-Hessian compressed core and benchmark one-metric observer package;

7. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Proposition 1 (What the preceding collapse theorem already proves)**.** *At current scope, the earlier current-image-core collapse theorem already proves all of the following.*

1. *The current body-response route* $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}$ *is exactly the quadratic response induced by* $S_\sigma^{\mathrm{disp,resp,vis}}$ *on the fixed section-centered displacement body.*

2. *The full canonical current-body image-core presentation is benchmark-facingly neutral once* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *is fixed.*

3. *All current benchmark-facing downstream uses of the present frontier factor through the visible-response datum* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$.

Proof. This is exactly the content of the earlier theorem collapsing the full canonical current-body image core to the canonical minimal visible-response representative and of its downstream factorization statement. □

3. Visible-Response-Faithful Benchmark Reductions

Definition 3 (Visible-response-faithful benchmark reduction)**.** *A* visible-response-faithful benchmark reduction *of the current section-centered hidden-response route is a reduction* $\mathfrak{B}_{\mathrm{vis}}$ *from deeper explicit substrate variables to the observer-facing sector such that, after admissible elimination of auxiliary or substrate-only variables and for each* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$,

1. *it preserves the fixed proto-section benchmark base of Definition 2(1);*

2. *it determines a benchmark-faithful candidate visible-response datum*

$$\mathfrak{D}_{\mathrm{vis}}(\mathfrak{B}_{\mathrm{vis}}, \sigma) = (\widetilde{\mathcal{D}}_\sigma^{\mathrm{disp,resp,vis}}, \widetilde{S}_\sigma^{\mathrm{disp,resp,vis}}),$$

   where

$$\widetilde{\mathcal{D}}_\sigma^{\mathrm{disp,resp,vis}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times \widetilde{V}_\sigma^{\mathrm{disp,resp,vis}}$$

   has fibers that are nonempty, compact, convex, contain 0*, and have nonempty interior, and*

$$\widetilde{S}_\sigma^{\mathrm{disp,resp,vis}}(y) : \widetilde{V}_\sigma^{\mathrm{disp,resp,vis}} \to \widetilde{V}_\sigma^{\mathrm{disp,resp,vis}}$$

   is a smooth positive self-adjoint family;

3. *it induces the quadratic visible-response law*

$$\widetilde{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y, v) \equiv \frac{1}{2}\left\langle v, \widetilde{S}_\sigma^{\mathrm{disp,resp,vis}}(y) v \right\rangle$$

   on $\widetilde{\mathcal{D}}_\sigma^{\mathrm{disp,resp,vis}}$*, and in the precise benchmark-facing sense of the current route this induced response reconstructs a benchmark-faithful current body-response core and hence the same current pullback body-response core, the same pullback body-operator core, the same pullback-operator core, the same pullback-quadratic core, the same section-centered residual-normal-form realization, the same pre-proto-section package, the same minimizer-section core, the same continuity-Hessian compressed core, and the same benchmark one-metric observer package;*

4. *it introduces no second operative metric, no enlarged low-energy matter law, and no change to the ordinary matter route.*

Remark 1. *Definition 3 is intentionally route specific. The theorem of this note concerns the active recorded section-centered hidden-response route, not every imaginable future redesign that changes the route itself.*

Definition 4 (Visible-response-faithful equivalence)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. Two candidate visible-response data*

$$(\mathcal{D}_\sigma^{(1)}, S_\sigma^{(1)}), \qquad (\mathcal{D}_\sigma^{(2)}, S_\sigma^{(2)})$$

over the fixed proto-section benchmark base are visible-response-faithfully equivalent *if, for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, there is a linear isomorphism*

$$\Phi_\sigma(y) : \mathrm{span}(\mathcal{D}_\sigma^{(1)}(y)) \xrightarrow{\cong} \mathrm{span}(\mathcal{D}_\sigma^{(2)}(y))$$

such that

$$\Phi_\sigma(y)(\mathcal{D}_\sigma^{(1)}(y)) = \mathcal{D}_\sigma^{(2)}(y)$$

and

$$\left\langle \Phi_\sigma(y)v, S_\sigma^{(2)}(y)\Phi_\sigma(y)v \right\rangle = \left\langle v, S_\sigma^{(1)}(y)v \right\rangle \qquad \textit{for every } v \in \mathcal{D}_\sigma^{(1)}(y).$$

Definition 5 (Minimal observer-relevant visible-response datum)**.** *For a visible-response-faithful benchmark reduction* $\mathfrak{B}_{\mathrm{vis}}$*, a datum* $\mathfrak{D}_{\mathrm{vis}}(\mathfrak{B}_{\mathrm{vis}})$ *is a* minimal observer-relevant visible-response datum *if:*

1. *it suffices to determine every benchmark-facing derivation statement available on the current section-centered hidden-response route;*
2. *no proper subdatum of it still suffices at that same benchmark-facing scope.*

4. Necessity of the Current Visible-Response Constituents

Proposition 2 (The displacement-body datum is necessary)**.** *Every minimal observer-relevant visible-response datum for a visible-response-faithful benchmark reduction must determine the section-centered displacement-body datum on the current route up to visible-response-faithful equivalence.*

Proof. The current route does not use an abstract positive operator with unspecified domain. It uses the recorded section-centered displacement body on which the visible quadratic response is actually realized. By Definition 2(2), that body records the admissible displacement fibers, contains the zero section, and has nonempty interior in each fiber. The fixed shell relation is likewise read on that body by the condition that $0 \in \mathcal{D}_\sigma^{\mathrm{ppresec}}(y)$ exactly on $\mathcal{Z}_\sigma^{\mathrm{psec}}$. If the displacement-body datum were omitted, neither the domain of the current visible-response law nor the shell discipline inherited by the current route would be fixed. Therefore no minimal benchmark-facing datum on this route may omit the displacement-body constituent. □

Proposition 3 (The visible-response operator is necessary)**.** *Every minimal observer-relevant visible-response datum for a visible-response-faithful benchmark reduction must determine the positive visible-response operator on the current route up to visible-response-faithful equivalence.*

Proof. The current route does not recover the current body-response core or the downstream benchmark-facing package from the displacement body alone. Proposition 1(1) uses the visible-response operator to recover the exact current body-response route by the quadratic formula

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y, v) = \frac{1}{2}\left\langle v, S_\sigma^{\mathrm{disp,resp,vis}}(y)v \right\rangle,$$

and Proposition 1(3) then carries all current downstream benchmark-facing uses through that recovery. If the operator constituent were omitted, the present route would no longer determine the current body-response core or the downstream package. Therefore no minimal benchmark-facing datum on this route may omit the operator constituent. □

Proposition 4 (The current visible-response datum is sufficient)**.** *The current canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *is sufficient to determine every benchmark-facing derivation statement available on the current section-centered hidden-response route.*

Proof. By Proposition 1(3), all current benchmark-facing downstream uses of the present frontier already factor through $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$. Therefore the current visible-response datum is sufficient at benchmark scope on the recorded route. □

5. Main Theorem

Theorem 1 (Canonical minimal visible-response representative necessity / uniqueness on the current route)**.** *For every visible-response-faithful benchmark reduction* $\mathfrak{B}_{\mathrm{vis}}$ *of the current section-centered hidden-response route, the canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *of Definition 2(5) is necessary and unique at benchmark scope up to visible-response-faithful equivalence. More precisely:*

1. *every minimal observer-relevant visible-response datum* $\mathfrak{D}_{\mathrm{vis}}(\mathfrak{B}_{\mathrm{vis}})$ *must determine both the section-centered displacement-body datum and the positive visible-response operator on the current route;*

2. $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *itself is sufficient to determine every benchmark-facing derivation statement available on that route;*

3. *any two minimal observer-relevant visible-response data are visible-response-faithfully equivalent to* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$*, and hence to each other.*

Proof. Item (1) is the combined content of Propositions 2 and 3. Those propositions show that no minimal observer-relevant datum on the current route may omit either the displacement-body constituent or the visible-response-operator constituent.

Item (2) is Proposition 4. The earlier current-image-core collapse theorem already proves that all current benchmark-facing downstream uses factor through $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$.

For item (3), let $\mathfrak{D}_{\mathrm{vis}}^{(1)}$ and $\mathfrak{D}_{\mathrm{vis}}^{(2)}$ be two minimal observer-relevant visible-response data for visible-response-faithful benchmark reductions of the current route. By item (1), each must determine the same kind of displacement-body and visible-response-operator constituents through which the route recovers the current body-response core and the benchmark-facing downstream package. By item (2), neither needs anything beyond those constituents at current benchmark scope. Therefore each is visible-response-faithfully equivalent to $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ in the sense of Definition 4. Hence they are visible-response-faithfully equivalent to each other as well. □

Corollary 1 (No smaller benchmark-facing datum remains on the current route)**.** *On the current section-centered hidden-response route, there is no genuinely smaller benchmark-facing datum below* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *up to visible-response-faithful equivalence.*

Proof. If a smaller datum existed, it would either omit the displacement-body constituent or the visible-response-operator constituent proved necessary by Propositions 2 and 3, or else merely rename the same constituents and therefore remain visible-response-faithfully equivalent to $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$. The first option contradicts necessity, and the second is not genuinely smaller at benchmark scope. □

Remark 2. *Corollary 1 is the benchmark-facing sense in which the current visible-response frontier is now exhausted internally on the recorded route. The route no longer has live benchmark-facing freedom in another same-visible-response restatement or in a re-expansion to a larger already-spent presentation.*

6. What Must Be Derived Next

Corollary 2 (Primary post-uniqueness burden)**.** *After Theorem 1, the next primary manuscript below the current frontier must do at least one of the following:*

1. *derive the current section-centered displacement-body datum from still more primitive explicit substrate structure rather than preserving it as already-recorded route data;*
2. *derive the current positive visible-response operator* $S_\sigma^{\mathrm{disp,resp,vis}}$ *from still more primitive explicit substrate structure rather than preserving it as already-recorded route data;*
3. *equivalently derive the current canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *directly from a smaller explicit substrate law without restoring the full current image-core presentation;*
4. *do one of the previous three while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.*

Another same-output deeper relay that merely preserves the same visible-response datum, or any re-expansion back to the full current image-core presentation, the larger reduced-carrier presentation, the older projection/body presentation, the full displacement-response substrate law, or the larger pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations, is not primary by default.

Proof. By Theorem 1 and Corollary 1, the current visible-response datum is already the unique minimal benchmark-facing datum on the recorded route up to visible-response-faithful equivalence. Therefore a new primary manuscript must derive that datum itself from still more primitive explicit substrate structure rather than merely restating it through another same-output presentation. □

Proposition 5 (The benchmark package remains fixed)**.** *The present necessity / uniqueness theorem preserves the benchmark one-metric package* (1)*. In particular, the operative metric remains exactly* $g^\sharp$*, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definitions 2(7) and 3(4). The present note removes only the remaining presentational freedom at current visible-response scope on the recorded route. □

7. Conclusion

The positive result of this note is narrower than a completed substrate derivation of GR but sharper than another routing reminder. On the current section-centered hidden-response route, the canonical minimal visible-response representative is now proved to be necessary and unique at benchmark scope up to visible-response-faithful equivalence.

That datum consists of the fixed section-centered displacement-body datum together with the positive visible-response operator realized on that body, under the same benchmark-faithfulness boundary. By the theorem of this note, no smaller genuinely benchmark-facing datum remains below it on the recorded route. The earlier current-image-core collapse theorem remains preserved main-line history because it first moved the live burden below the full current image-core presentation. The present necessity / uniqueness theorem then spends the remaining presentational freedom at that smaller frontier.

This sharpens the honest next burden. A new primary manuscript must no longer merely show that an even deeper substrate layer reproduces the same visible-response datum or re-expands to a larger already-spent presentation. It must derive or uniquely force the current visible-response datum itself from still more primitive explicit substrate structure while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.